

WEEK 1

Team Based Learning

iRAT activities

Questions with wrong answers: NONE

TA signature:

WEEK 2

Git

iRAT activities

Questions with wrong answers: NONE

Important notes

Git commands

Commands	Functions
cd "folder_name"	Move from one folder to another
wq	Save your changes and quit
w	Save the file
q	Quit Vim
q!	Quit without saving
i	Enter Insert Mode
Esc	Leave Insert mode and go back to Command mode
git clone	Create a local copy of an existing remote Git repository
git add	Saves changes from your working directory into Git staging area (index)
git commit	Captures a snapshot of your project's currently staged changes and saves it permanently to your local repository history
git push	Command uploads local repository commits to a remote server
git pull	Downloads changes from a remote repository and immediately integrates them into your current local branch
git init	Creates a new, empty Git repository or reinitialises an existing one

git fetch	Downloads commits, files, and references from a remote repository into your local repository without merging them into your active local branches
git log	Displays a repository's complete commit history in reverse chronological order
git status	Displays the current state of your working directory and the staging area, showing which changes are tracked, untacked, or ready to be committed
git fork	A server-side, independent copy of a repository hosted under your own account on platforms (But you can't edit it, it's just a copy)

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WEEK 3

Classes and Objects

iRAT activities

Questions with wrong answers

Question 7

Complete

Marked out of 1.00

 Flag question

What is printed by the following code?

```
public class Person {
    private String name;
    public Person(String name) {
        name = name;
    }
    public String getName() {
        return name;
    }
}
```

```
Person p = new Person("Alice");
System.out.println(p.getName());
```

Select one:

- ☒ a. Alice
- ☐ b. null
- ☐ c. Compilation error
- ☐ d. No output — the program crashes with an exception
- ☐ e. "

Correct answers: B: null

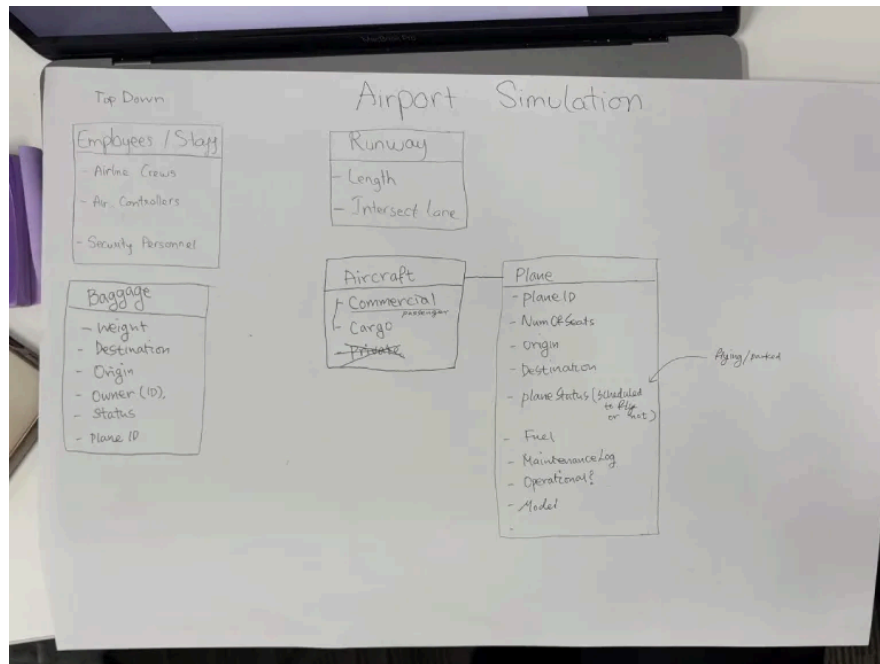
Reason: Because there is no “this.name=name” therefore java would print out “null” as “name=name”

Project

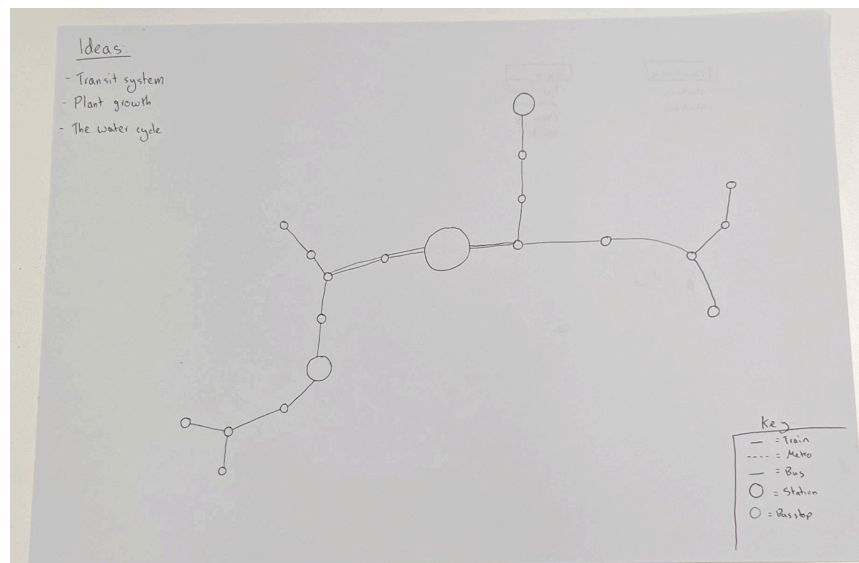
Draw.io link:

https://app.diagrams.net/#G1_CHOv7-QWzYlbBotdgZEz5XpHFofoiV#%7B%22pageId%22%3A%22Vp9QY6zV6l614nxmznQm%22%7D

Our group's ideas of simulation: Airport simulation



Other group's ideas simulation: Traffic simulation



TA signature: minh

WEEK 4

Inheritance and Overloading

iRAT activities

Questions with wrong answers

Question 8

Complete

Marked out of 1.00

Flag question

Given the following code, which statement is true about the constructor call `super()`?

```
class Animal {
    String name;
    Animal() {
        name = "unknown";
    }
}
class Dog extends Animal {
    String breed;
    Dog(String breed) {
        this.breed = breed;
    }
}
```

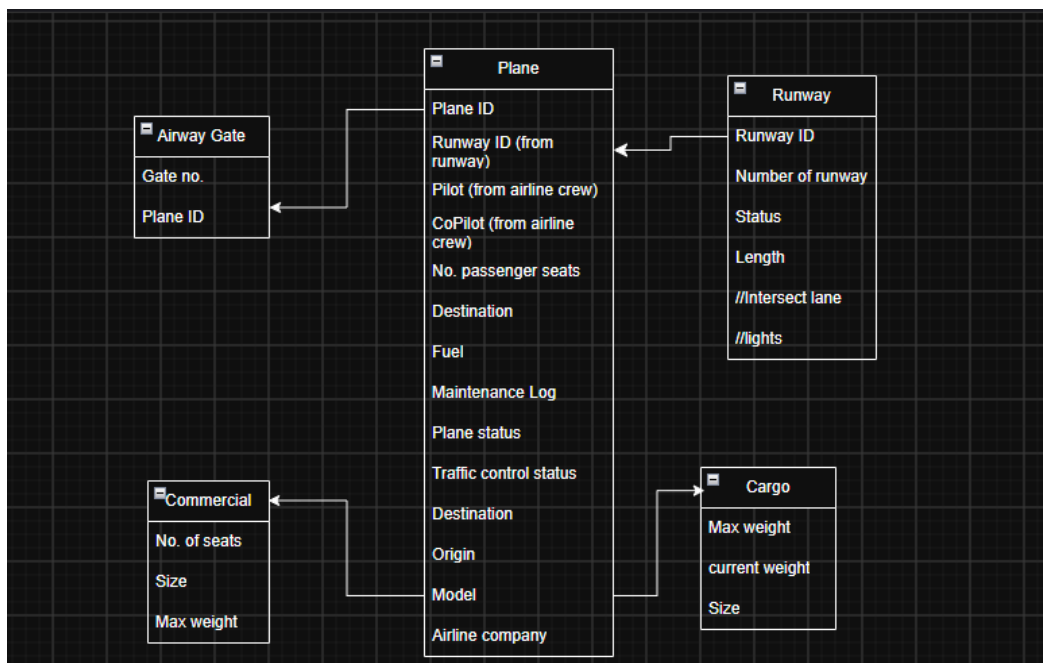
Select one:

- ☐ a. Dog's constructor cannot call `super()` because Dog has its own fields.
- ☐ b. The code does not compile because Dog must explicitly call `super()`.
- ☐ c. The compiler implicitly inserts `super()` as the first statement in Dog's constructor.
- ☒ d. The `Animal()` constructor is not called because Dog does not reference name.
- ☐ e. `super()` must be the last statement in Dog's constructor to ensure breed is initialised first.

Correct answers: C

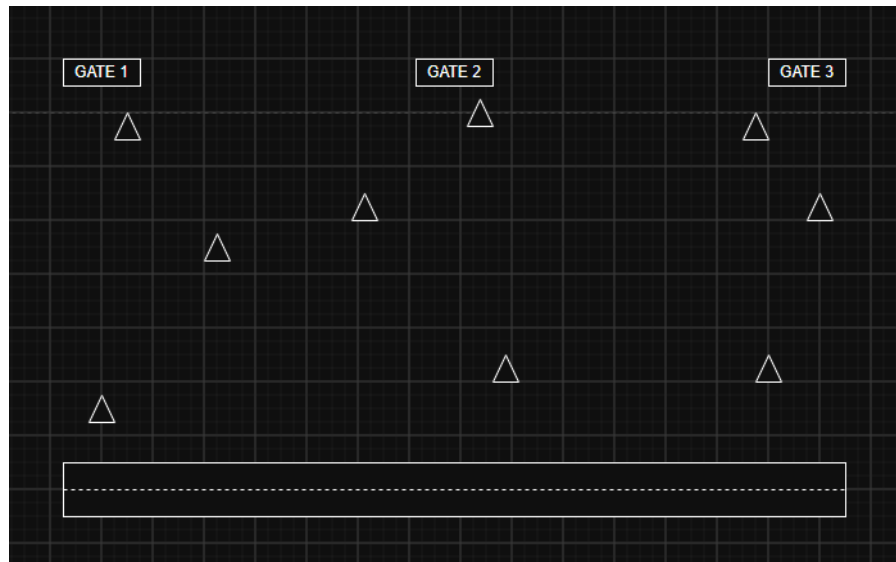
Reason:

Our group's ideas of simulation: Airport Traffic Simulation



- We have decided to focus on the plane and runway classes for now as it helps the most with our simulation.
- Compared to last week where we have lots of classes and entities, we removed baggage, staff, aircrews and passengers out of it.
- For now we could have 1 run way, 3 gates and a few planes (triangles) in the panel to move around. For example:

My Ideas for later simulation



- Maybe for later weeks, we could develop a simulation more on passengers getting on the plane once the plane stops.
- Or a bus that drives passengers to the plane on a bus lane.

TA signature: minh

WEEK 5

Generics

iRAT activities

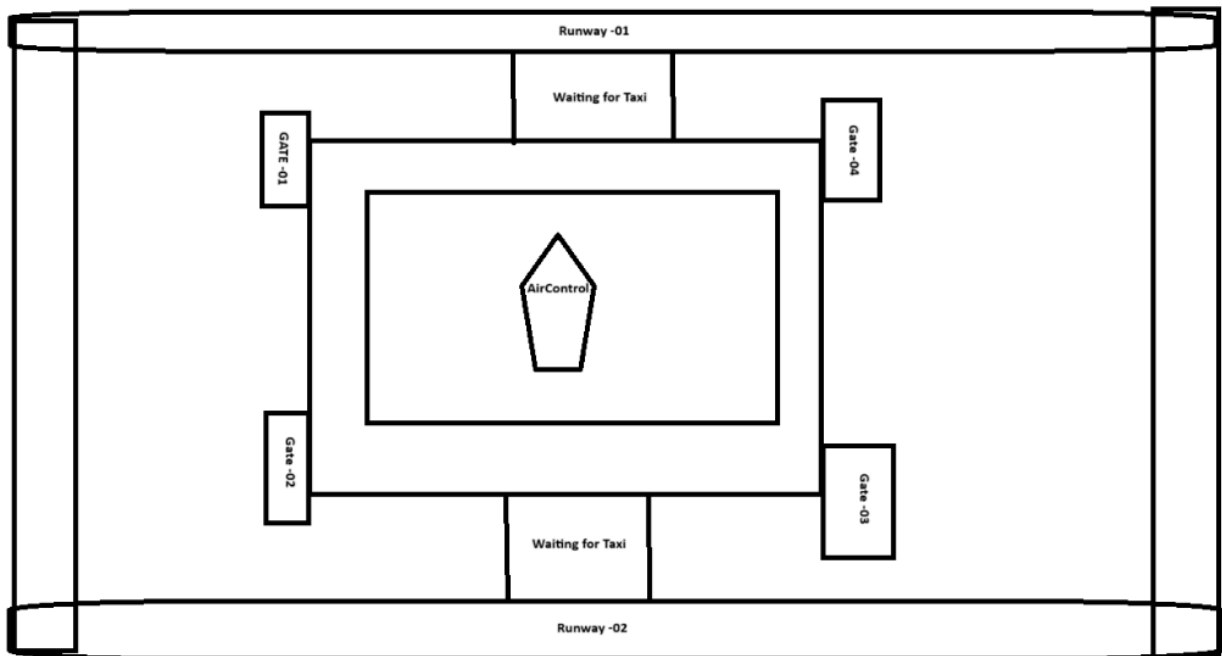
Questions with wrong answers

Correct answers:

Reason:

Project

Airport sketch



- Start putting on setters and getters

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WEEK 6

Exceptions

iRAT activities

Questions with wrong answers: None

Important notes

Try and Catch:

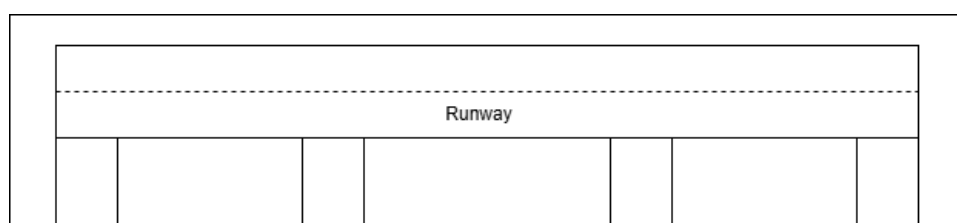
- Basically “run this program and try this, if it doesn’t work, catch this exception without crashing the program.”
- Useful because: things can go wrong while a program runs, and you want your program to handle them sensibly.
 - For example: “Enter your age:” but users enter a String “hello”, the program will then run “please enter valid age” instead of crashing.

Finally block:

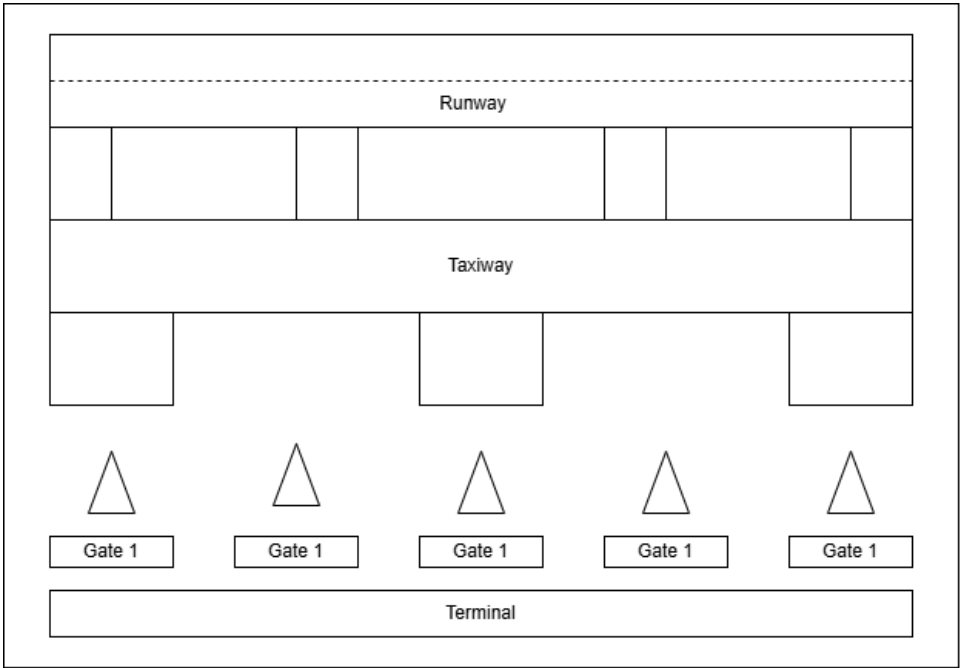
- Run whether an exception happens or not.
- Useful because: finally block still runs even if there’s an error.

NullPointerException	Unchecked exception	No
IOException	Checked exception	Yes
ArrayIndexOutOfBoundsException	Unchecked exception	No
IllegalArgumentException	Unchecked exception	No
StackOverflowError	Unchecked exception	No

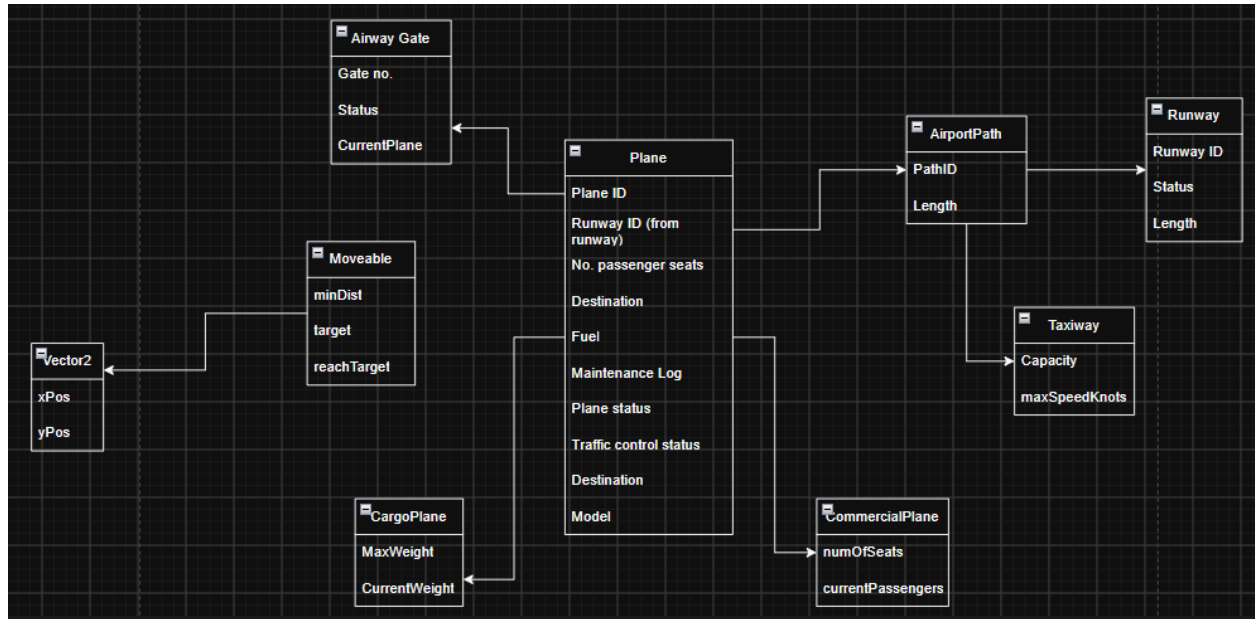
My ideas for later simulation



My Group’s ideas for later simulation



UML design



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WEEK 7 Patterns 1 PRESENTATION

Presentation Feedback:

- Organise our files
- Sort out a way to spawn planes
- Develop better coloring design for our panel

TA signature:

